



PLVNet: EEG-based trust classification using Phase Locking Value connectivity and deep neural networks

Julakha Jahan Jui^{a,*}, Imali T. Hettiarachchi, Asim Bhatti, Douglas Creighton

Institute for Intelligent Systems Research and Innovation (IISRI), Deakin University, Waurn Ponds, Geelong, 3216, Victoria, Australia

ARTICLE INFO

Keywords:

Trust in automation
EEG
Functional connectivity
Phase Locking Value (PLV)
Deep learning
PLVNet
Neural dynamics
Human–automation interaction
Brain–computer interface (BCI)
Cognitive monitoring

ABSTRACT

Trust in automation is critical for effective human–automation interaction, yet traditional subjective measures are limited in capturing rapid and dynamic changes in user trust. This study introduces PLVNet, a novel deep neural network architecture designed to classify trust versus distrust states from EEG functional connectivity features derived using Phase Locking Value (PLV). PLV features were extracted across six canonical EEG frequency bands (Delta, Theta, Alpha, Beta, Low Gamma, High Gamma) from 30-channel EEG recordings. The PLVNet model was evaluated using three complementary approaches: aggregated analysis (5× 5 stratified cross-validation), participant-wise analysis, and leave-one-subject-out (LOSO) cross-validation. PLVNet significantly outperformed convolutional neural network (CNN), support vector machine (SVM) and k-nearest neighbours (KNN) classifiers across all evaluation schemes. Beta and Low Gamma bands provided the highest discriminative power, while functional connectivity analysis revealed that trust is associated with enhanced fronto-parietal and fronto-occipital synchronisation, reflecting global network integration, whereas distrust shows fragmented connectivity patterns. PLVNet's ability to capture non-linear inter-dependencies in connectivity patterns highlights its advantages over conventional methods. These findings demonstrate that PLV-based connectivity robustly reflects trust-related neural dynamics, underscoring the potential of PLVNet for real-time, objective monitoring of trust in human–automation systems, which paves the way for adaptive and neuro-aware interfaces.

1. Introduction

Trust plays a critical role in human–automation interaction (HAI), influencing decision-making, collaboration, and overall system performance [1–3]. In high-stakes domains such as autonomous driving, aviation, and industrial automation, inappropriate levels of trust can have severe consequences [3]: under-trust may lead to system underuse, while over-trust can result in misuse and reduced situational awareness [4]. Therefore, accurate and real-time assessment of trust is essential for enabling safe and efficient human–automation teaming. Recently, neurometric assessments using electroencephalography (EEG) have gained traction for tracking trust dynamics, particularly in applications such as autonomous driving and adaptive automation [5–7].

Traditional trust assessment techniques, including self-report questionnaires [8] and behavioural observations [9], provide useful insights but suffer from key limitations: they are subjective, retrospective, and poorly suited to capturing rapid, moment-to-moment fluctuations in trust [7]. EEG, in contrast, offers a direct, objective, and temporally

precise measure of neural activity, enabling continuous monitoring of cognitive and affective states in real time [10].

Neuroscientific evidence indicates that trust and distrust involve coordinated activity across distributed brain regions, manifesting as distinct EEG patterns, including event-related potentials [11], oscillatory dynamics [12], and changes in functional connectivity across key networks during trust evaluations [13]. In particular, functional connectivity within the fronto-parietal network, particularly in the theta and upper-alpha bands, has been widely associated with higher-order cognitive integration and executive control processes [14], which are critical during trust-related evaluations. Moreover, oscillatory activity in other frequency bands also contributes significantly to differentiating neural states of trust and distrust.

- Alpha power supports the maintenance of internal cognitive states, such as engagement and sustained trust [15].
- Theta activity in prefrontal regions is associated with adaptive control and responses to uncertainty [16].

* Corresponding author.

E-mail address: jjui@deakin.edu.au (J.J. Jui).

<https://doi.org/10.1016/j.combiomed.2025.111269>

Received 11 October 2025; Received in revised form 24 October 2025; Accepted 28 October 2025

Available online 3 November 2025

0010-4825/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

- Beta-band synchronisation has been linked to top-down regulation and cognitive set maintenance during trustful interactions [17].

Importantly, stress also modulates trust-related neural dynamics by altering oscillatory patterns and functional connectivity. Elevated stress typically increases theta and beta activity in frontal regions, reflecting heightened cognitive control demands, while simultaneously reducing alpha power, which may impair the stability of sustained trust evaluations [18,19].

To characterise these neural dynamics, a wide range of EEG feature extraction techniques has been developed. Frequency-domain features, such as power spectral density (PSD) [10], quantify oscillatory energy and are widely applied in cognitive-state classification. Nonlinear features, including recurrence quantification analysis (RQA) [20] and entropy-based measures [21], capture the complexity and stability of brain dynamics.

Functional connectivity metrics measure synchronisation between distributed cortical regions and provide insights into large-scale network coordination. Given that trust involves distributed, network-level integration, connectivity-driven approaches may offer a promising pathway for improved trust quantification. However, in EEG-based trust recognition, most prior studies have primarily focused on time-domain and frequency-domain features [10,22], with limited exploration of phase-based functional connectivity.

Recent studies highlight the value of connectivity approaches. Huang et al. [13] applied dynamic causal modelling (DCM) to EEG data and demonstrated distinct effective connectivity patterns between high-trust and low-trust conditions, indicating that directed inter-regional influences vary systematically with trust levels. More recently, Bales et al. [23] introduced inter-channel network metrics, showing that network-level synchronisation dynamics can more accurately track trust in human-autonomy interactions than conventional signal-power features. These findings underscore the potential of connectivity-based analyses to capture the distributed neural mechanisms underlying trust.

Building on these insights, this study proposes a novel framework, PLVNet, that integrates phase-locking value (PLV)-based functional connectivity with a lightweight deep learning (DL) model for EEG-based trust classification. PLV is employed to capture stable phase synchronisation between brain regions while being robust to amplitude variations, making it particularly suitable for multi-trial and real-time scenarios [18,19]. The proposed PLVNet architecture is designed to automatically learn discriminative connectivity patterns from full PLV matrices, enabling accurate classification of trust and distrust states. By introducing this framework, we aim to bridge the gap between traditional EEG features and advanced connectivity-driven DL approaches for trust recognition. To systematically evaluate the effectiveness of PLVNet and the underlying neural features, the following research questions guide this study:

1. Can the proposed DL model, PLVNet, reliably learn discriminative representations from PLV matrices for accurate trust classification?
2. Can PLV-based functional connectivity features effectively characterise neural dynamics underlying trust and distrust in HAI?
3. Which PLV frequency bands serve as the most informative neural markers for trust-related processes?
4. Which PLV connectivity features and brain regions contribute most to trust-related neural processes, and how consistent are these patterns across participants?

The remainder of this paper is organised as follows. Section 2 details the materials and methods, including participant recruitment, task design, experimental setup, EEG data preprocessing, and data segmentation. This section also describes the feature extraction procedures, including PLV computation and feature construction, followed

by an overview of the proposed PLVNet deep learning model, its architecture, and the training and evaluation protocols. Section 3 presents the experimental findings, covering classification performance across all participants, participant-wise and frequency band-wise analyses, and functional connectivity patterns associated with trust and distrust states. Section 4 provides a comprehensive discussion of the results, integrating neurophysiological interpretations, practical implications, study limitations, and directions for future research. Finally, Section 5 concludes the paper.

2. Materials and methods

2.1. Participants

A total of twelve eligible male participants; age range: 20–45 years; mean age: 29.85 ± 5.45 years) were recruited through group email invitations and poster advertisements. All participants were physically healthy, reported no known anxiety or stress disorders, and had no history of neurological conditions or cognitive ailments (e.g., epilepsy, cataplexy) or heart and circulatory diseases. Additional exclusion criteria included any impairment of physical or mental health that could affect task performance, any condition that would prevent sitting comfortably for up to 1.5 h.

To control for physiological factors that might influence experimental measures, participants were instructed to refrain from intense physical exercise and alcohol consumption for 24 h before their session, maintain a normal sleep routine the night before, and avoid sleep deprivation. They were also asked to abstain from consuming food, tea, coffee, or other caffeinated beverages for one hour before the experiment.

All participants provided written informed consent before commencing the study and were informed that their participation was voluntary, their data would remain confidential, and they could withdraw at any time without penalty. The study protocol received approval from the Deakin University Human Research Ethics Committee (Reference Number: SEBE-2024-17).

2.2. Task design

The experiment employed a word-elicitation task [8,24] specifically designed to evoke cognitive and emotional states associated with trust and distrust. Unlike prior studies [24], which used stand-alone words or generic cues, the present paradigm incorporated contextually enriched scenarios paired with each trust-related word. This design enhancement aimed to facilitate deeper cognitive appraisal and emotional resonance, thereby eliciting more robust and meaningful neural responses [25]. Furthermore, neurophysiological evidence supports the validity of word-based paradigms in capturing genuine trust-related neural processes. Oh et al. [26] demonstrated that EEG signals elicited by trust and distrust words exhibit distinct oscillatory patterns corresponding to each condition, while Firoz et al. [24] reported that pseudo-real-time, word-based trust tasks produce neural signatures consistent with real-world trust dynamics. These findings collectively affirm that contextually guided word-elicitation tasks can reliably evoke neural activity comparable to interactive trust paradigms.

Building on this rationale, participants were presented with 30 stimuli, comprising 15 trust-related and 15 distrust-related words. Each word was paired with a descriptive scenario crafted to prompt the recollection of personal experiences relevant to trust or distrust. For example, the trust word *assurance* was paired with the scenario: “Imagine you’re on a long road trip, and you’re relying on a navigation app to guide you through unfamiliar areas. As you drive, the app consistently provides real-time traffic updates, rerouting you around roadblocks and heavy traffic. You’ve been using this app for years, and it has never let you down. You feel completely assured that it will get you to your destination on time, just like it always has”. Such scenario-word pairs

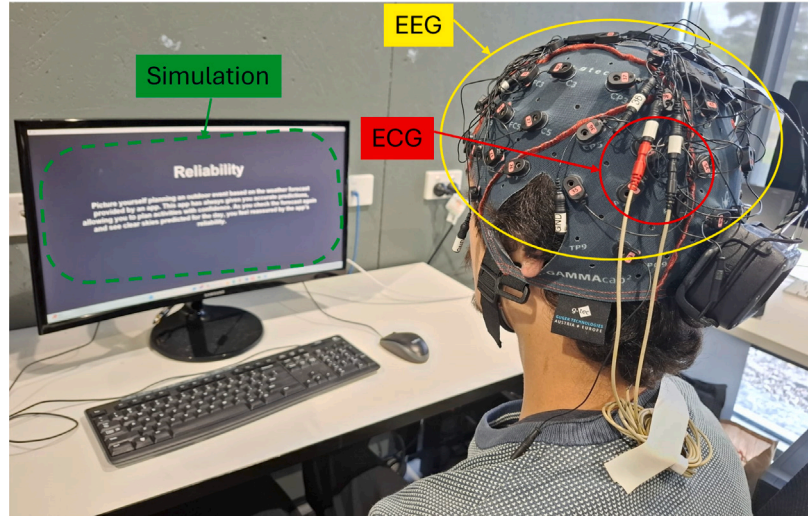


Fig. 1. Experimental setup. A subject is wearing an EEG cap while viewing the computer screen during a word elicitation task.

were designed to immerse participants and elicit the targeted cognitive state for accurate neural measurement.

Participants were instructed to focus on the presented word and scenario attentively and to mentally recall any personal event that resonated with the provided context. They were encouraged to reflect freely, without suppressing any emotions or thoughts that emerged during the task, thereby simulating real-world situations where trust-related cognition and affect often occur concurrently.

Each stimulus was displayed for 55 s, providing ample time for participants to engage with the content and reflect on relevant memories or emotions. A 5-second inter-stimulus interval was included to allow mental resetting before the onset of the next stimulus.

2.3. EEG acquisition setup

EEG and electrocardiogram (ECG) data were collected during the experiments, but for this study, only EEG recordings are analysed and discussed further. The EEG data were collected via a 64-channel wireless g.Nautilus active electrode multi-purpose bio-signal acquisition system (g.tec medical engineering GmbH, Austria). EEG was recorded from 32 selected scalp channels at a sampling rate of 250 Hz, following the international 10–20 electrode placement system. The selection of these 32 channels (FP1, FP2, AF3, AF4, F7, F3, FZ, F4, F8, FC5, FC1, FC2, FC6, T7, C3, CZ, O2, C4, T8, CP5, CP1, CP2, CP6, P7, P3, PZ, P4, P8, PO3, PO4, O1, OZ) was based on their relevance to cognitive and affective processing, particularly in the context of trust-related neural activity. Fig. 1 shows the experimental setup.

2.4. EEG data preprocessing

EEG data underwent a structured preprocessing pipeline to enhance signal quality and remove artifacts. Initially, two channels (CP6 and P7) were excluded from all recordings due to consistently corrupted signals resulting from faulty electrodes. Baseline correction was performed by subtracting the mean value from each channel to eliminate the DC offset. The data were then filtered using a 0.5 Hz high-pass filter to remove slow drifts and a 50 Hz low-pass filter to retain physiologically relevant oscillations, including high gamma activity. To eliminate power line interference, a notch filter targeting the 49–51 Hz range was applied [27]. These preprocessing steps ensured that the neural signals retained frequency components of interest (delta through high gamma) while minimising noise, thereby improving the reliability of downstream analysis.

Subsequently, the EEG signals were refined using the automatic and tunable artifact removal (ATAR) algorithm [28], with the tunable parameter set to $\beta = 0.6$. The ATAR algorithm, based on Wavelet Packet Decomposition, operates on individual EEG channels to suppress artifacts such as muscle movements or eye blinks while preserving meaningful neural activity. The selection of $\beta = 0.6$ was based on an empirical evaluation of a range of values (0.01 to 0.1 in steps of 0.02, and 0.1 to 1 in steps of 0.1), balancing artifact suppression and signal preservation.

Following artifact correction, the EEG signals were normalised (z-score transformation) to standardise the amplitude across channels and participants. This step reduced inter-individual variability and facilitated fair comparison in downstream analyses. The pseudocode representation in Algorithm 1 provides a step-by-step description of this preprocessing workflow.

Algorithm 1: EEG Preprocessing Pipeline for Trust/Distrust Quantification

Input: Raw EEG signals $X \in \mathbb{R}^{C \times T}$ from 30 channels
Output: Cleaned and normalised EEG signals $\hat{X}$

Step 1: Channel Rejection;
 $X \leftarrow X \setminus \{\text{CP6, P7}\};$

Step 2: Baseline Correction;
for each channel $c \in X$ **do**
 $x_c(t) \leftarrow x_c(t) - \mu_c;$ // μ_c is mean of channel c
end

Step 3: Filtering;
Apply 0.5 Hz high-pass filter: $x_c(t) \leftarrow \text{HPF}(x_c(t));$
Apply 50 Hz low-pass filter: $x_c(t) \leftarrow \text{LPF}(x_c(t));$
Apply notch filter (49–51 Hz): $x_c(t) \leftarrow \text{Notch}(x_c(t));$

Step 4: Artifact Correction (ATAR);
for each channel $c \in X$ **do**
 Decompose $x_c(t)$ using wavelet packet decomposition;
 Suppress artifact coefficients with parameter $\beta = 0.6$;
 $x_c(t) \leftarrow$ Reconstruct signal from modified coefficients;
end

Step 5: Normalisation;
for each channel $c \in X$ **do**
 $z_c(t) \leftarrow \frac{x_c(t) - \mu_c}{\sigma_c};$ // z-score normalisation
end

Step 6: Quality Check;
Visualise raw vs. corrected signals to confirm noise suppression;
return $\hat{X} = \{z_c(t)\};$

2.5. Data segmentation

To enable accurate and temporally resolved analysis of trust-related EEG dynamics, the normalised data were segmented in a two-stage process. Each trial had a duration of 55 s and was sampled at 250 Hz,

resulting in 13,750 data points per trial. Subsequently, each trial was divided into non-overlapping 5-second windows, yielding 11 windows per trial. This process generated a total of 3960 segments across all trials and participants, with 1980 segments each for the trust and distrust classes.

2.6. Feature extraction

Functional connectivity features were extracted using the Phase Locking Value (PLV), a widely used measure of phase synchronisation between EEG channels. PLV quantifies the consistency of phase differences across time, making it particularly suitable for capturing neural coupling associated with trust-related cognitive states.

2.6.1. Phase locking value (PLV) computation

PLV between two EEG channels x and y was computed as:

$$PLV_{xy} = \left| \frac{1}{N} \sum_{k=1}^N e^{i(\phi_x(k) - \phi_y(k))} \right|, \quad (1)$$

where N is the number of samples in a segment, and $\phi_x(k)$ and $\phi_y(k)$ are the instantaneous phases of channels x and y at sample k , obtained via the Hilbert transform. The resulting PLV values range from 0 (no phase synchronisation) to 1 (perfect synchronisation).

2.6.2. Feature construction

PLV was computed for all unique channel pairs from the 30 EEG channels, resulting in:

$$\binom{30}{2} = 435 \text{ channel pairs.}$$

To capture frequency-specific connectivity patterns, PLV was computed across six canonical frequency bands: **Delta** (0.5–4 Hz), **Theta** (4–8 Hz), **Alpha** (8–13 Hz), **Beta** (13–30 Hz), **Low Gamma** (30–45 Hz), and **High Gamma** (45–50 Hz).

2.7. Proposed PLVNet deep learning model

To classify trust and distrust states from PLV-based functional connectivity features, we developed a lightweight fully connected neural network, referred to as *PLVNet*. The motivation for this design arises from two key challenges inherent in EEG-based trust analysis: (1) the high dimensionality of connectivity features and (2) the limited number of labelled samples per participant. Conventional approaches that directly operate on adjacency matrices, such as image-based connectivity networks or 1D convolutional neural networks (CNNs), are prone to over-fitting in such scenarios because they either process overly large input spaces or attempt to learn local temporal patterns that may not capture global connectivity dynamics relevant to trust.

Each 5-second EEG segment was transformed into a 2,610-dimensional PLV feature vector (435 channel pairs $\times$ 6 frequency bands). To reduce redundancy and improve generalisation, a non-parametric Mann–Whitney U test [29] was applied independently within each training fold to select features that significantly differed between trust and distrust conditions ($p < 0.001$), thereby avoiding peeking bias. This test was chosen because it does not assume normality, making it suitable for EEG-derived PLV features, which often exhibit non-Gaussian distributions. Feature stability was assessed by examining their frequency of selection across folds, revealing a consistent core set of connections in frontal, parietal, and fronto-central regions, supporting reproducibility and robust model inputs. On average, 92 out of 2,610 features ($\approx 3.5\%$) were retained per fold, representing the most discriminative connectivity patterns between trust and distrust conditions.

The PLVNet architecture comprises three fully connected hidden layers with 256, 128, and 64 units, respectively. Batch normalisation is applied after each layer to stabilise training and accelerate

convergence, while ReLU activations introduce non-linearity. Dropout regularisation ($0.3 \rightarrow 0.3 \rightarrow 0.2$) is employed after each layer to further reduce overfitting. The final output layer consists of two units with softmax activation, providing class probabilities for *trust* and *distrust*. Unlike CNNs, which learn sequential patterns but require larger datasets, PLVNet directly captures global connectivity-based representations from flattened feature vectors, making it both lightweight and effective for high-dimensional, small-sample EEG data.

Training was performed using the Adam optimiser with categorical cross-entropy loss. To prevent overfitting and improve generalisation, early stopping was employed by monitoring the validation loss, halting training if no improvement was observed over a patience of 5 epochs. The complete pipeline for PLVNet-based trust quantification, including EEG segmentation, PLV computation, fold-wise feature selection, model training with early stopping, and evaluation, is illustrated in Algorithm 2. This pipeline ensures robust, generalisable classification and highlights the advantages of PLVNet:

1. Efficient feature utilisation: By selecting only statistically significant PLV features, the model avoids redundant and noisy inputs.
2. Robust performance: Fold-wise feature selection combined with a lightweight architecture reduces over-fitting and improves generalisation.
3. Computational efficiency: The fully connected design with moderate hidden layer sizes allows rapid training and real-time applicability.
4. Direct global connectivity modelling: PLVNet captures global synchronisation dynamics that are directly relevant to trust, unlike models that focus on local temporal patterns.

Algorithm 2: PLVNet Workflow for EEG-Based Trust/Distrust Classification with Trial-wise Grouping

Input: EEG data for each participant, labels (*trust/distrust*)

Output: Predicted trust/distrust labels, performance metrics

```

for each participant  $p$  do
  for each trial  $i$  do
    Segment EEG into 5-second windows;
    Compute PLV for all 30 channels across 6 frequency bands;
    Flatten PLV matrices into a feature vector of length 2610;
    // Keep all segments from the same trial together for fold assignment
  for each of 5 runs do
    for each of 5 cross-validation folds do
      Assign trials (all their segments) to the training or test set to prevent inter-trial leakage;
      Apply uni-variate statistical feature selection ( $t$ -test or Mann–Whitney U) on training data;
      Retain features with  $p < 0.001$ ;
      // Feature selection is fold-specific to avoid using test data
      Initialise PLVNet:
        • Fully connected layers: Input  $\rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow$  Output (2 classes)
        • Batch normalisation applied after each hidden layer
        • ReLU activation after each hidden layer
        • Dropout applied after each hidden layer:  $0.3 \rightarrow 0.3 \rightarrow 0.2$ 
      Train PLVNet on training data with:
        • Optimiser: Adam
        • Loss: Categorical cross-entropy
        • Epochs: 30, Batch size: 64
        • Early stopping monitored on validation loss with patience = 5
      Evaluate PLVNet on the test set;
      Compute metrics: Accuracy, Precision, Recall, F1-score, ROC-AUC;
    Aggregate results across folds and runs;
    Visualise performance: confusion matrices, ROC curves;
  
```

Table 1Classification performance of PLVNet and benchmark models (mean $\pm$ std) based on 5×5 evaluation.

Model	Accuracy	Precision	Recall	F1-score	AUC-ROC
PLVNet	0.9207 $\pm$ 0.0110	0.9191 $\pm$ 0.0198	0.9233 $\pm$ 0.0185	0.9209 $\pm$ 0.0109	0.9779 $\pm$ 0.0050
CNN	0.9066 $\pm$ 0.0170	0.9060 $\pm$ 0.0222	0.9080 $\pm$ 0.0296	0.9066 $\pm$ 0.0174	0.9685 $\pm$ 0.0090
SVM	0.8843 $\pm$ 0.0155	0.8772 $\pm$ 0.0184	0.8940 $\pm$ 0.0208	0.8854 $\pm$ 0.0154	0.9532 $\pm$ 0.0104
KNN	0.8578 $\pm$ 0.0194	0.8463 $\pm$ 0.0222	0.8748 $\pm$ 0.0229	0.8602 $\pm$ 0.0188	0.9288 $\pm$ 0.0151

2.8. PLVNet model training and evaluation

The PLVNet model was implemented in Python (version 3.11) using the PyTorch library and trained on a workstation equipped with a 13th Gen Intel Core i9-13900HX CPU (2.2 GHz) and an NVIDIA GeForce RTX 4080 Laptop GPU with CUDA 12.2 and cuDNN 8.9. Training was performed with a batch size of 64 over 30 epochs, optimising the balance between computational efficiency and model performance.

To ensure a rigorous assessment of generalisability, a 5-fold cross-validation strategy was employed and repeated across 5 independent runs. For each evaluation metric, including classification accuracy, precision, recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (AUC-ROC), the mean and standard deviation (std) were computed and reported as mean $\pm$ std. In addition, 95% confidence intervals (CI) and effect sizes (Cohen's d) were calculated to provide statistical context for the performance differences. Confusion matrices are also presented to visualise classification outcomes. This comprehensive evaluation enables robust performance estimation while accounting for variability due to data partitioning and random initialisation, and provides statistical evidence of model superiority.

All models received the same subset of discriminative PLV features after feature selection, which was performed separately within each training fold to prevent information leakage. To further mitigate temporal dependencies and inter-trial correlations, all segments from the same trial were kept together within a fold, and leave-one-subject-out (LOSO) validation was employed for participant-level evaluation. This ensures that no data from the same trial or participant appears in both training and test sets, preventing temporal and inter-trial leakage and improving the robustness of performance estimates [30].

For benchmarking, PLVNet's performance was compared against three alternative models: a CNN with three fully connected layers, a support vector machine (SVM) with an RBF kernel, and a k-nearest neighbours (KNN) classifier. While the CNN mirrored PLVNet's architecture in terms of layer dimensions ($256 \rightarrow 128 \rightarrow 64$ units), ReLU activations, and dropout (0.3–0.2), key differences exist in their structural design and training mechanism. PLVNet incorporates additional batch normalisation layers after each fully connected layer, which stabilises training, accelerates convergence, and improves generalisation, particularly in small, noisy EEG datasets. Moreover, PLVNet was trained with careful regularisation and dropout scheduling, which enhances its ability to capture subtle functional connectivity patterns from PLV features.

All models received the same subset of discriminative PLV features after feature selection, which was performed separately within each training fold to prevent information leakage. Hyperparameters for each model were optimised within the training folds: SVM used an RBF kernel with C and gamma selected via grid search, and KNN's k was optimised from 3 to 15 using Euclidean distance. All models were trained and evaluated using the same subject-independent cross-validation protocol, providing a fair comparison of their ability to classify trust and distrust states based on PLV-derived connectivity features. Classical ML models such as SVM and KNN were included to provide interpretable baselines and to demonstrate the added value of PLVNet in capturing complex connectivity patterns from EEG data.

To statistically evaluate differences in model performance across participants, a Friedman test was conducted for each evaluation metric (accuracy, precision, recall, F1-score, and AUC-ROC). The Friedman test is a non-parametric test suitable for repeated-measures data and

does not assume normality, making it appropriate for participant-wise EEG classification results [31]. When the Friedman test indicated significant differences, post-hoc pairwise comparisons were performed using the Wilcoxon signed-rank test with Bonferroni correction to control for multiple comparisons [31].

3. Experimental findings

3.1. Classification performance of PLVNet and benchmark models (combined participants)

To assess population-level generalisability, EEG data from all 12 participants were aggregated to form a single dataset. This approach evaluates each model's ability to handle diverse neural patterns, reflecting real-world variability across individuals. As shown in Table 1, PLVNet consistently outperformed the benchmark models, demonstrating superior discrimination between *trust* and *distrust* states. Its high classification performance across multiple metrics indicates that PLVNet effectively captures subtle, distributed connectivity patterns, supporting its reliability and robustness for generalisable trust classification in HAI contexts.

Further insights are provided in Table 2, which reports 95% confidence intervals (CI) and Cohen's d effect size ranges for the classification performance of PLVNet and benchmark models. PLVNet consistently exceeded the comparative models across all key metrics, with narrower confidence intervals indicating stable performance across cross-validation folds. The large Cohen's d values underscore a substantial and meaningful improvement over baseline methods, reflecting strong discriminative capability.

In comparison to CNN, SVM, and KNN, PLVNet demonstrated superior central tendency and reduced variability, confirming its robustness in modelling phase synchrony from EEG connectivity data. Collectively, these results validate the effectiveness of PLVNet's architecture in capturing non-linear neural dependencies, establishing it as a reliable framework for trust classification in HAI scenarios.

In addition to classification accuracy, computational efficiency was compared between PLVNet and the CNN model (Table 3). PLVNet demonstrated a significant advantage in efficiency, requiring fewer parameters (104,386 vs. 248,258) and lower computational cost (105.28 vs. 279.43 KMac FLOPs). Moreover, PLVNet achieved an average inference time of 0.047 ms per sample, approximately 3.5 times faster than the CNN baseline. This efficiency highlights the model's suitability for real-time or embedded EEG-based trust recognition systems, where low-latency processing is essential.

To provide further insight into model predictions, confusion matrices for PLVNet, CNN, SVM, and KNN are presented in Fig. 2. These matrices illustrate the distribution of true and predicted *trust* and *distrust* states, highlighting PLVNet's superior classification capability. Furthermore, Fig. 3 shows the ROC curves for all four models. The ROC curves provide a comprehensive view of the trade-off between true positive rates and false positive rates, demonstrating that PLVNet achieves the highest AUC, followed by SVM, CNN, and KNN. This confirms that PLVNet consistently outperforms the benchmark models in discriminating between *trust* and *distrust* states.

To assess whether the observed differences in classification performance were statistically significant, a Friedman test was conducted for each evaluation metric (accuracy, precision, recall, F1-score, and AUC-ROC) across the four models. The Friedman test indicated significant differences for all metrics:

Table 2

95% confidence intervals (CI) and Cohen's d effect size ranges for the classification performance of PLVNet and benchmark models. Cohen's d values represent the effect size of PLVNet relative to each benchmark model across all performance metrics.

Model	Accuracy CI	Precision CI	Recall CI	F1-score CI	AUC-ROC CI	Cohen's d range
PLVNet	0.9161–0.9253	0.9109–0.9272	0.9157–0.9310	0.9164–0.9254	0.9758–0.9800	36.78–187.14
CNN	0.8995–0.9136	0.8969–0.9152	0.8957–0.9202	0.8994–0.9138	0.9648–0.9722	18.74–85.98
SVM	0.8779–0.8907	0.8696–0.8848	0.8855–0.9026	0.8790–0.8917	0.9489–0.9575	42.97–91.55
KNN	0.8498–0.8658	0.8371–0.8555	0.8654–0.8843	0.8524–0.8679	0.9225–0.9350	38.07–61.35

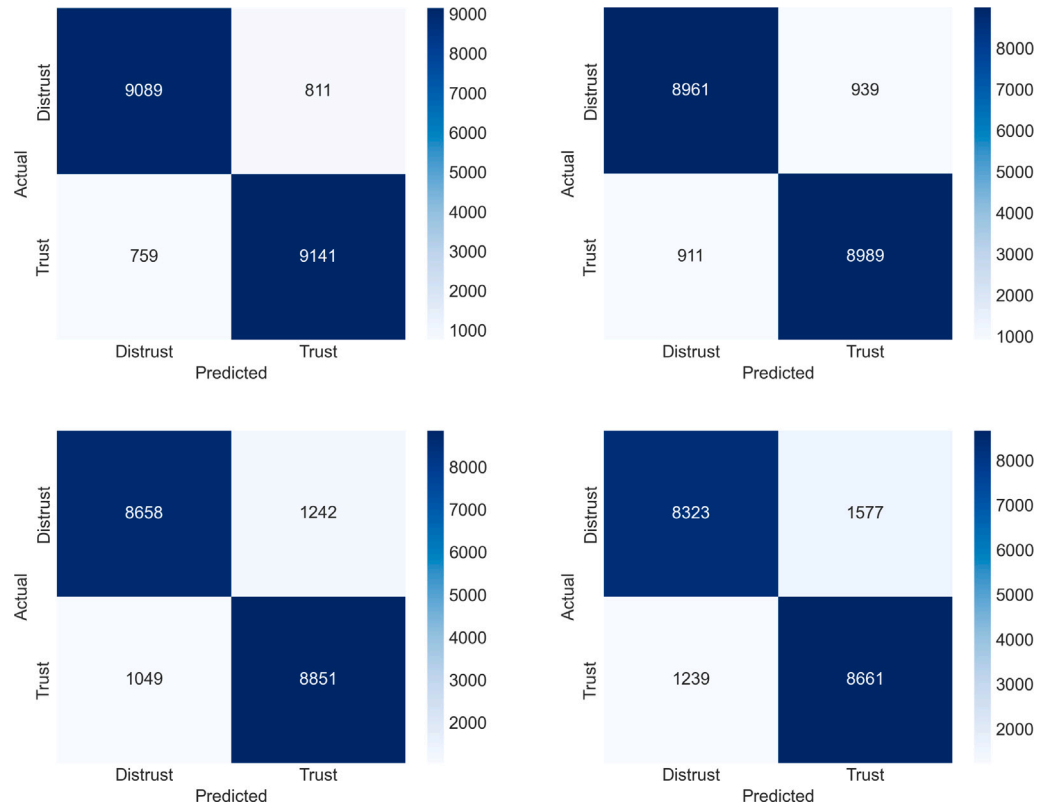


Fig. 2. Aggregated confusion matrices of the four models for combined participant data. Top-left: PLVNet; Top-right: CNN; Bottom-left: SVM; Bottom-right: KNN. Each matrix shows the distribution of true versus predicted *trust* and *distrust* states.

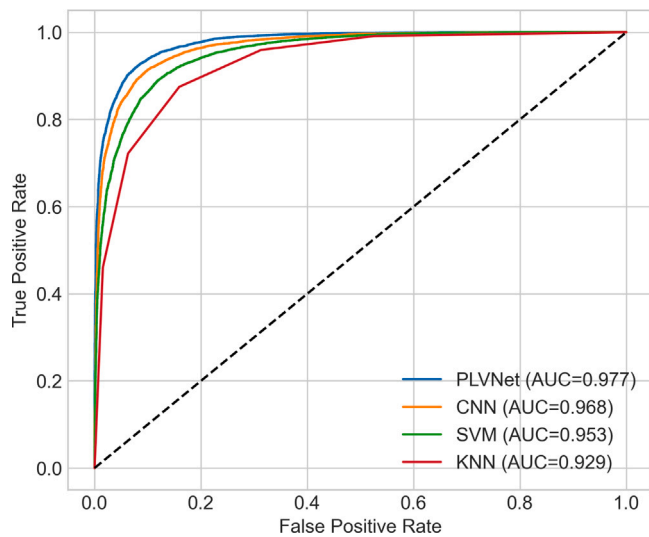


Fig. 3. Comparison of ROC curves for PLVNet, CNN, SVM, and KNN using combined participant data. PLVNet achieves the highest AUC, demonstrating superior discrimination between *trust* and *distrust* states.

Table 3

Computational efficiency of PLVNet and CNN models.

Model	Parameters	FLOPs (KMac)	Inference time (ms)
PLVNet	104,386	105.28	0.047
CNN	248,258	279.43	0.169

- **Accuracy:** $\chi^2 = 61.25$, $p = 2.11 \times 10^{-13}$
- **Precision:** $\chi^2 = 49.36$, $p = 8.27 \times 10^{-11}$
- **Recall:** $\chi^2 = 25.42$, $p = 1.04 \times 10^{-5}$
- **F1-score:** $\chi^2 = 61.05$, $p = 2.37 \times 10^{-13}$
- **AUC-ROC:** $\chi^2 = 68.52$, $p = 9.72 \times 10^{-15}$

Post-hoc pairwise Wilcoxon signed-rank tests with Bonferroni correction were performed to identify specific differences between models, as presented in Table 4. The results indicate that PLVNet significantly outperformed KNN across all metrics and also achieved significant improvements over CNN and SVM for most metrics, including accuracy, F1-score, and AUC-ROC (all adjusted $p < 0.05$).

These results confirm that PLVNet provides robust discrimination between *trust* and *distrust* states compared to benchmark models, highlighting its reliability across multiple evaluation metrics. The consistent superiority of PLVNet, particularly in AUC-ROC (where it significantly

Table 4

Summary of significant pairwise differences (Wilcoxon signed-rank test, Bonferroni corrected) for aggregated results. A checkmark indicates that PLVNet significantly outperforms the benchmark model (adjusted $p < 0.05$). Adjusted p-values are reported for all comparisons.

Metric	PLVNet > CNN	PLVNet > SVM	PLVNet > KNN
Accuracy	✓ (5.10×10^{-3})	✓ (1.00×10^{-4})	✓ (1.00×10^{-4})
Precision	2.53e-01	✓ (1.00×10^{-4})	✓ (1.00×10^{-4})
Recall	2.88e-01	✓ (2.00×10^{-4})	✓ (1.00×10^{-4})
F1-score	✓ (2.00×10^{-3})	✓ (1.00×10^{-4})	✓ (1.00×10^{-4})
AUC-ROC	✓ (1.00×10^{-4})	✓ (1.00×10^{-4})	✓ (1.00×10^{-4})

Table 5

Participant-wise classification accuracy (mean $\pm$ std) for PLVNet and benchmark models based on 5×5 evaluation using updated feature selection threshold ($p < 0.001$).

Participant	PLVNet	CNN	SVM	KNN
P1	0.996 $\pm$ 0.008	0.993 $\pm$ 0.012	0.998 $\pm$ 0.006	0.948 $\pm$ 0.027
P2	0.992 $\pm$ 0.014	0.985 $\pm$ 0.012	0.990 $\pm$ 0.011	0.970 $\pm$ 0.022
P3	0.925 $\pm$ 0.038	0.920 $\pm$ 0.033	0.923 $\pm$ 0.033	0.870 $\pm$ 0.042
P4	0.945 $\pm$ 0.028	0.916 $\pm$ 0.034	0.921 $\pm$ 0.028	0.879 $\pm$ 0.039
P5	0.898 $\pm$ 0.038	0.786 $\pm$ 0.055	0.862 $\pm$ 0.047	0.811 $\pm$ 0.044
P6	0.992 $\pm$ 0.011	0.988 $\pm$ 0.014	0.987 $\pm$ 0.010	0.963 $\pm$ 0.021
P7	0.984 $\pm$ 0.012	0.979 $\pm$ 0.014	0.982 $\pm$ 0.016	0.978 $\pm$ 0.015
P8	0.961 $\pm$ 0.039	0.945 $\pm$ 0.030	0.969 $\pm$ 0.021	0.935 $\pm$ 0.029
P9	0.788 $\pm$ 0.081	0.702 $\pm$ 0.064	0.779 $\pm$ 0.042	0.701 $\pm$ 0.043
P10	0.988 $\pm$ 0.015	0.985 $\pm$ 0.016	0.995 $\pm$ 0.010	0.973 $\pm$ 0.022
P11	0.958 $\pm$ 0.025	0.927 $\pm$ 0.027	0.956 $\pm$ 0.020	0.903 $\pm$ 0.036
P12	0.972 $\pm$ 0.021	0.964 $\pm$ 0.027	0.968 $\pm$ 0.022	0.959 $\pm$ 0.020

outperformed all benchmarks), underscores its strong discriminative capability for trust classification tasks (see Table 4).

3.2. Participant-wise classification performance

This subsection presents a participant-wise analysis, where each model was trained and tested individually for each participant. Personalised evaluation is crucial given the high inter-subject variability in EEG signals, which can significantly impact the performance of real-world trust assessment or fatigue detection systems. To capture these individual differences, classification accuracy was computed separately for each participant. Table 5 summarises the accuracy of PLVNet and benchmark models (CNN, SVM, and KNN) across all participants using a 5×5 evaluation, providing a clear comparison of model performance while highlighting both overall trends and participant-specific variability.

Overall, PLVNet maintained strong and consistent performance across participants, achieving accuracies above 0.95 for participants P1, P6, P7, P8, P10, and P12, demonstrating its solid generalisation capability. Although CNN and SVM showed comparable or slightly higher accuracies for certain participants (e.g., P2, P3, P4, P7 and P8), PLVNet remained stable and competitive across the cohort. KNN continued to yield the lowest and most variable results, particularly for participants P1, P3, and P9, reinforcing the advantage of PLVNet's connectivity-based feature representation in capturing trust-related neural dynamics.

Fig. 4 illustrates participant-wise precision and recall with std across all participants. Precision reflects the model's ability to correctly identify trust trials, while recall measures the proportion of correctly identified positive trials. PLVNet consistently achieves higher precision and recall compared to CNN, SVM, and KNN, with particularly strong results for participants P1, P9, P10, P11 and P12. The error bars represent variability across 5-fold cross-validation, averaged over 5 independent runs. Across participants, the trends in precision closely mirror those observed in recall, further reinforcing the robustness and generalisability of the proposed model. Overall, PLVNet demonstrates

Table 6

Participant-wise Wilcoxon signed-rank post-hoc test results for classification accuracy. A checkmark indicates that PLVNet significantly outperforms the benchmark model (adjusted $p < 0.05$). Adjusted p-values are reported for all comparisons.

Participant	PLVNet > CNN	PLVNet > SVM	PLVNet > KNN
P1	5.73e-01	1.41e-01	1.00e+00
P2	1.00e+00	1.00e+00	✓(1.52e-04)
P3	1.00e+00	1.00e+00	✓(1.52e-04)
P4	✓(7.58e-03)	✓(7.28e-04)	✓(8.33e-05)
P5	1.00e+00	1.00e+00	1.00e+00
P6	1.00e+00	✓(2.42e-04)	✓(8.33e-05)
P7	1.00e+00	1.00e+00	✓(1.02e-04)
P8	1.00e+00	1.00e+00	✓(5.33e-15)
P9	1.00e+00	✓(5.44e-08)	✓(5.33e-15)
P10	✓(3.96e-03)	1.00e+00	✓(3.29e-08)
P11	✓(1.56e-02)	✓(1.22e-08)	✓(3.31e-09)
P12	1.00e+00	✓(1.30e-05)	✓(5.17e-07)

a reliable and subject-independent performance, maintaining high accuracy, precision, and recall even for participants with more variable responses.

To determine whether differences in classification performance between models were statistically significant, an overall Friedman test was first conducted across all participants for each evaluation metric, complementing the aggregated-level analyses reported in the previous section. The results indicated a significant effect of model choice for all metrics:

- **Accuracy:** $\chi^2 = 23.45$, $p = 2.4 \times 10^{-5}$
- **Precision:** $\chi^2 = 15.62$, $p = 1.3 \times 10^{-3}$
- **Recall:** $\chi^2 = 17.80$, $p = 5.6 \times 10^{-4}$
- **F1-score:** $\chi^2 = 22.75$, $p = 3.1 \times 10^{-5}$
- **ROC-AUC:** $\chi^2 = 21.90$, $p = 6.8 \times 10^{-5}$

These findings indicate that, across participants, model choice had a statistically significant impact on performance, consistent with the trends observed in the aggregated-level results.

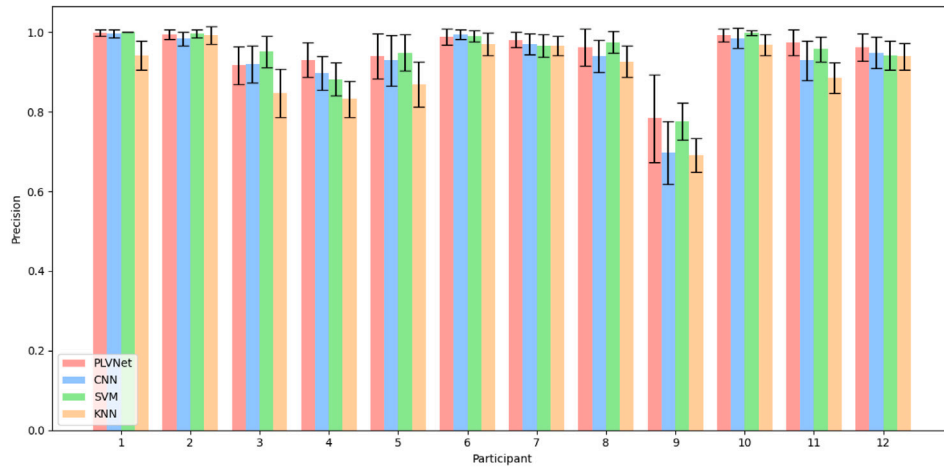
To examine model performance at the individual participant level, post-hoc pairwise comparisons were performed using the Wilcoxon signed-rank test with Bonferroni correction. Table 6 summarises these results, where a checkmark indicates that PLVNet significantly outperforms a benchmark model for a given participant (adjusted $p < 0.05$). The corresponding adjusted p-values are provided in parentheses to convey the strength of evidence for each comparison. This approach enables clear and transparent reporting of both statistical significance and the magnitude of the observed effects.

The results in Table 6 show that PLVNet demonstrates superior performance over benchmark models for many participants. Significant improvements are observed over KNN for 10 out of 12 participants, and over SVM for 5 out of 12 participants. While comparisons with CNN show fewer significant differences (3 out of 12 participants), PLVNet demonstrates competitive performance across all participants. These findings underscore the robustness and participant-level effectiveness of PLVNet in accurately distinguishing trust states.

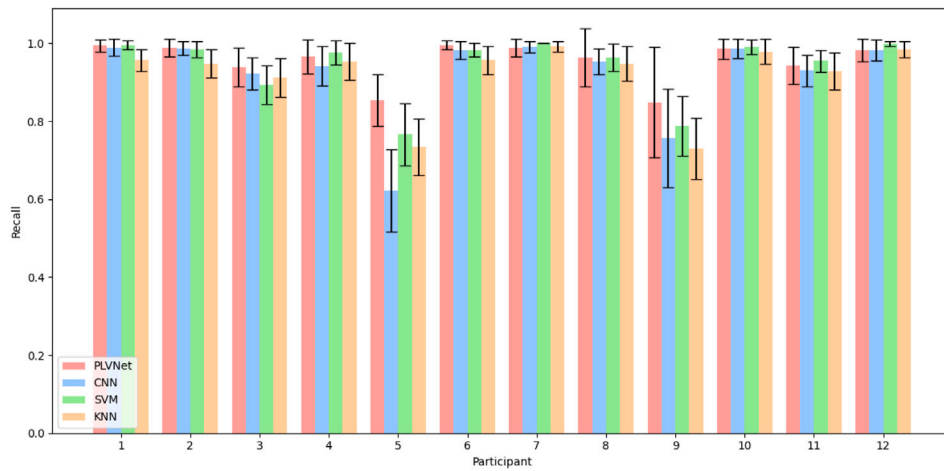
Similar participant-wise post-hoc analyses were performed for precision, recall, F1-score, and ROC-AUC, revealing a consistent pattern of superior performance by PLVNet across participants and metrics. These results demonstrate that the observed differences in model performance are not only statistically significant at the group level but are also robust at the individual participant level.

3.3. Performance evaluation using LOSO cross-validation

LOSO cross-validation was employed to evaluate the models' ability to generalise to unseen individuals. In this validation approach, one participant's data is held out for testing while the model is trained on



(a) Participant-wise precision of all models.



(b) Participant-wise recall of all models.

Fig. 4. Comparison of participant-wise classification performance. Precision (a) indicates the proportion of correctly predicted trust trials, and recall (b) represents correctly identified positive trials across participants.

data from the remaining participants. This procedure is repeated iteratively until each participant has served as the test subject. LOSO provides a stringent assessment of model robustness in real-world scenarios where subject-specific data may be unavailable during training.

Table 7 summarises participant-wise classification accuracy under the LOSO cross-validation scheme for PLVNet and three benchmark models (CNN, SVM, and KNN). PLVNet achieved the highest accuracy for most participants, highlighting its ability to generalise across individuals. Although CNN and SVM slightly outperformed PLVNet for a small number of participants (e.g., P2, P5, and P9), PLVNet consistently maintained the highest mean accuracy overall, demonstrating superior robustness and reliability compared to the benchmark models.

The narrower variability observed in PLVNet's performance indicates stronger robustness and stability across participants, reflecting its superior capability in capturing non-linear EEG connectivity patterns related to *trust* states. Overall, these results highlight the effectiveness of PLVNet for participant-independent trust classification in EEG-based applications.

To determine whether the differences in classification performance under LOSO cross-validation were statistically significant, a Friedman test was conducted across all participants. The results indicated a significant effect of model choice on accuracy ($\chi^2 = 27.30$, $p = 5.09 \times 10^{-6}$), confirming that model selection substantially impacts participant-level performance.

Post-hoc Wilcoxon signed-rank tests with Bonferroni correction were conducted to evaluate participant-wise differences between PLVNet and benchmark models under LOSO cross-validation. As shown in Table 8, PLVNet significantly outperformed KNN across all participants and SVM for most participants (8 out of 12). Comparisons with CNN revealed significant improvements for a majority of participants (8 out of 12), while non-significant differences corresponded to participants where CNN slightly exceeded or closely matched PLVNet accuracy. Overall, these results confirm that PLVNet consistently provides superior accuracy at the individual participant level, highlighting its robustness and generalisability to unseen participants in diverse EEG datasets.

3.4. Frequency band-wise classification performance

To identify which EEG frequency bands carry the most discriminative information for trust and distrust classification, PLVNet was evaluated separately on each of the six canonical frequency bands: Delta (0.5–4 Hz), Theta (4–8 Hz), Alpha (8–13 Hz), Beta (13–30 Hz), Low Gamma (30–45 Hz), and High Gamma (45–50 Hz). Table 9 summarises the results across all participants and runs.

The results indicate that Low Gamma and Beta bands exhibit the highest classification performance, with accuracy above 91%–92% and ROC-AUC above 97%–98%. Alpha shows moderate predictive power

Table 7
Participant-wise classification accuracy under LOSO cross-validation.

Participant	PLVNet	CNN	SVM	KNN
P1	0.7104	0.6713	0.6628	0.6512
P2	0.7212	0.7328	0.7335	0.6845
P3	0.7023	0.6645	0.6812	0.6630
P4	0.7325	0.6891	0.6776	0.6524
P5	0.6824	0.7026	0.6712	0.6598
P6	0.7118	0.6830	0.6664	0.6547
P7	0.7034	0.6619	0.6482	0.6365
P8	0.7285	0.6914	0.6838	0.6621
P9	0.6938	0.7116	0.6952	0.6685
P10	0.7220	0.6815	0.6724	0.6560
P11	0.7028	0.6668	0.6612	0.6409
P12	0.7336	0.7031	0.6840	0.6652
Mean $\pm$ SD	0.7131 $\pm$ 0.0178	0.6880 $\pm$ 0.0188	0.6848 $\pm$ 0.0185	0.6554 $\pm$ 0.0162

Table 8

Participant-wise post-hoc Wilcoxon signed-rank test results for LOSO accuracy. A checkmark indicates that PLVNet significantly outperforms the benchmark model for that participant (adjusted $p < 0.05$). Adjusted p-values are reported in parentheses.

Participant	PLVNet > CNN	PLVNet > SVM	PLVNet > KNN
P1	✓(4.00e−03)	✓(2.00e−03)	✓(1.00e−04)
P2	1.40e−01	✓(1.80e−02)	✓(1.20e−03)
P3	✓(1.00e−02)	✓(6.00e−03)	✓(2.00e−04)
P4	✓(3.00e−03)	✓(4.00e−03)	✓(1.00e−04)
P5	9.00e−02	✓(2.50e−03)	✓(4.00e−04)
P6	✓(1.80e−02)	✓(1.50e−03)	✓(2.00e−04)
P7	✓(1.20e−02)	6.50e−02	✓(3.00e−04)
P8	✓(9.00e−03)	8.00e−02	✓(2.00e−04)
P9	1.10e−01	✓(2.10e−03)	✓(1.00e−04)
P10	✓(5.00e−03)	5.80e−02	✓(2.00e−04)
P11	✓(7.00e−03)	✓(1.20e−03)	✓(1.00e−04)
P12	9.50e−02	✓(2.80e−03)	✓(2.00e−04)

(72% accuracy, 81% AUC), while High Gamma demonstrates intermediate performance (83% accuracy, 92% AUC). Delta and Theta bands show lower performance (accuracy 66%–69%, AUC 73%–77%), suggesting limited contribution to trust/distrust discrimination in PLV-based connectivity.

Notably, feature selection using the Mann–Whitney U test ($p < 0.001$) resulted in substantial dimensionality reduction across all frequency bands: Beta (94.8% reduction), Low Gamma (95.2% reduction), High Gamma (94.5% reduction), Alpha (96.8% reduction), Theta (96.1% reduction), and Delta (95.7% reduction). Despite this aggressive feature pruning, Beta and Low Gamma bands maintained exceptional performance, indicating that these frequency ranges contain highly discriminative and robust connectivity patterns for trust classification.

These findings suggest that PLV connectivity patterns in the Low Gamma and Beta frequency ranges are the most informative for classifying trust and distrust states, supporting targeted feature selection in future work. Subsequent analyses will examine participant-wise band performance and precision/recall trends.

3.5. Functional connectivity representation

To complement the classification results and provide a neurophysiological interpretation of discriminative features, PLV connectivity patterns were analysed for trust and distrust states. Functional connectivity matrices were computed for the Beta (13–30 Hz) and Low-Gamma (30–45 Hz) bands, which demonstrated the highest classification performance. This analysis enabled the identification of network-level connectivity patterns associated with varying trust levels, highlighting the neurophysiological correlates underlying the observed behavioural states.

The results reveal a systematic reorganisation of brain networks across multiple perspectives (Figs. 5 and 6). Trust was characterised

by a widespread increase in global functional connectivity. In the Beta band, 126 connections showed significant differences, with 92 (73.0%) exhibiting higher PLV during trust compared to distrust. In the Low-Gamma band, 178 connections were significant, with 143 (80.3%) showing enhanced synchronisation during trust. This global integration is visually apparent in the difference matrices (Figs. 5c and 6c), which are dominated by red hues, particularly in the Low-Gamma band, indicating stronger connectivity during trust.

Scalp topography plots (Figs. 5d and 6d) illustrate that trust is associated with dense long-range fronto-parietal, fronto-occipital, and frontal-central connectivity (e.g., Fp1–Pz, Fp2–CP2, Fp1–FC1). Extensive connections between prefrontal and posterior electrodes suggest that trust engages a distributed network supporting higher-order social cognition, executive control, and sensory integration.

Conversely, distrust is associated with more localised functional disconnection. The most consistent decreases are observed in temporal and central regions (approximated by electrodes T7/T8, FC1/FC2), which exhibit reduced synchronisation with parietal and occipital areas (e.g., T7–CP5, T7–O1; $\Delta = -0.015$ to -0.034 across bands). These sparser connectivity patterns (Figs. 5d and 6d) suggest disruption in neural circuits supporting theory of mind, attentional control, and sensory integration.

Frequency-specific analysis indicates that the Low-Gamma band exhibits both more extensive and stronger increases in connectivity during trust-related tasks compared to the Beta band. This is evident both in the number of significant connections and the visual density of networks in scalp topography. These results reinforce the classification findings (Table 9) and highlight the critical role of high-frequency oscillatory networks in supporting complex cognitive computations underlying trust-based decisions.

Neurophysiologically, gamma-band oscillations are associated with local and long-range cortical synchronisation, attentional integration, and binding of distributed neural representations. During trust, enhanced Low-Gamma connectivity may reflect rapid coordination of cognitive and affective processes, including risk assessment, social evaluation, and decision-making. This aligns with prior findings showing increased gamma-band connectivity during tasks involving trust decisions, underscoring its role in social cognition [32].

These findings are further illustrated in the brain-wide PLV connectivity diagrams (Fig. 7), which visualise the functional connections for both trust (Figs. 7a and 7b) and distrust (Figs. 7c and 7d) states in the Beta and Low-Gamma bands. In these diagrams, the top 50% of strongest functional connections are displayed for each state. Consistent with the statistical results, trust is characterised by enhanced fronto-parietal and fronto-occipital synchronisation (Figs. 7a and 7b), reflecting greater large-scale network integration. In contrast, distrust exhibits more fragmented connectivity, with fewer long-range connections, particularly along temporal-parietal pathways (Figs. 7c and 7d). Notably, the centro-parietal nodes (CP1, Pz, CP2, and Cz) consistently show the strongest connectivity across both frequency bands and states, suggesting that these regions may act as critical hubs for information integration during trust-related decision-making.

Table 9PLVNet classification performance across individual EEG frequency bands (mean $\pm$ std).

Frequency band	Accuracy	Precision	Recall	F1-score	ROC AUC
Delta	0.6669 $\pm$ 0.0173	0.6688 $\pm$ 0.0190	0.6633 $\pm$ 0.0400	0.6653 $\pm$ 0.0221	0.7374 $\pm$ 0.0199
Theta	0.6892 $\pm$ 0.0142	0.6861 $\pm$ 0.0200	0.7008 $\pm$ 0.0429	0.6923 $\pm$ 0.0188	0.7706 $\pm$ 0.0150
Alpha	0.7232 $\pm$ 0.0247	0.7167 $\pm$ 0.0273	0.7408 $\pm$ 0.0487	0.7275 $\pm$ 0.0288	0.8138 $\pm$ 0.0218
Beta	0.9058 $\pm$ 0.0147	0.9080 $\pm$ 0.0263	0.9053 $\pm$ 0.0449	0.9054 $\pm$ 0.0170	0.9742 $\pm$ 0.0065
Low Gamma	0.9172 $\pm$ 0.0118	0.9119 $\pm$ 0.0194	0.9244 $\pm$ 0.0222	0.9177 $\pm$ 0.0118	0.9772 $\pm$ 0.0048
High Gamma	0.8278 $\pm$ 0.0177	0.8332 $\pm$ 0.0329	0.8229 $\pm$ 0.0381	0.8269 $\pm$ 0.0182	0.9203 $\pm$ 0.0129

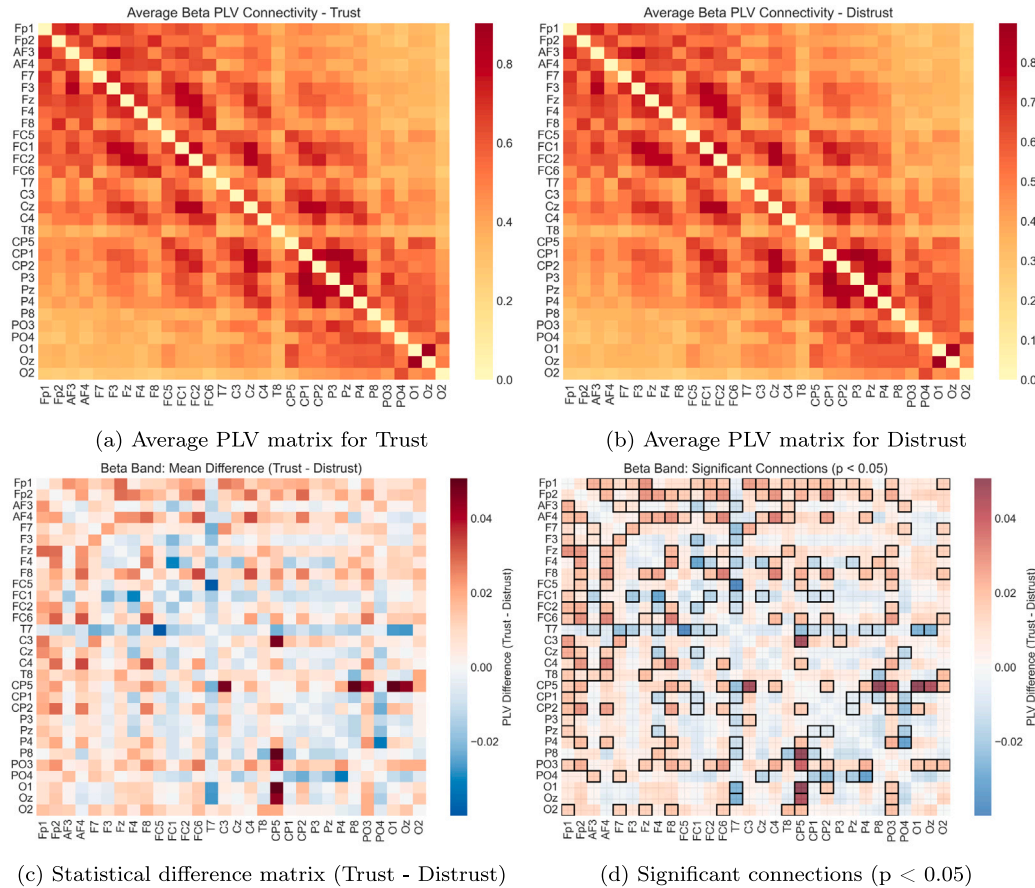


Fig. 5. Functional connectivity analysis of trust and distrust states in the Beta band (13–30 Hz). (a) Average PLV connectivity pattern during trust states, showing enhanced synchronisation across multiple brain regions. (b) Average PLV connectivity pattern during distrust states, demonstrating more fragmented and reduced connectivity. (c) Statistical difference matrix highlighting connections that are significantly stronger (red) or weaker (blue) during trust compared to distrust. (d) Significant connections ($p < 0.05$) after thresholding by PLV strength, emphasising the most discriminative network patterns for trust classification. **Colour scale:** Blue indicates lower PLV values, red indicates higher PLV values. In the statistical difference matrices, red indicates connections stronger during Trust than Distrust, and blue indicates connections weaker during Trust than Distrust.

3.6. Feature importance and neural contribution

To identify the most robust neural markers for trust classification, we analysed PLV features that demonstrated consistent discriminative power across all cross-validation folds. [Table 10](#) presents the PLV features that were selected in 100% of cross-validation folds ($p < 0.001$, Mann–Whitney U test), highlighting their exceptional stability and reliability for trust-distrust discrimination.

To visualise the broader set of discriminative connectivity features, the top 20 PLV features with the highest statistical significance ($p < 0.001$ from the previously applied Mann–Whitney U test) were selected for presentation in [Fig. 8](#). These features span multiple brain regions, including frontal, central, parietal, and occipital areas, and are colour-coded by frequency bands. Notably, high and low gamma bands are prominently represented, reflecting coordinated oscillatory activity associated with cognitive processes underpinning trust.

Table 10

Most stable PLV features for trust-distrust classification.

Feature	Frequency band	Selection rate	Brain regions
CP5_O1	High Gamma	100%	Parietal-Occipital
CP5_P8	High Gamma	100%	Parietal
CP5_O1	Low Gamma	100%	Parietal-Occipital
CP5_O1	Beta	100%	Parietal-Occipital
CP5_PO3	Low Gamma	100%	Parietal
FP2_CP2	Low Gamma	100%	Frontal-Central
FP2_PO3	Low Gamma	100%	Frontal-Parietal
FP2_PO3	High Gamma	100%	Frontal-Parietal
CP5_P8	Theta	100%	Parietal
FP2_F4	High Gamma	100%	Frontal

Note: Features selected in 100% of cross-validation folds ($n = 25$).

The stability analysis ([Table 10](#)) reveals several key patterns that complement the broader feature importance shown in [Fig. 8](#):

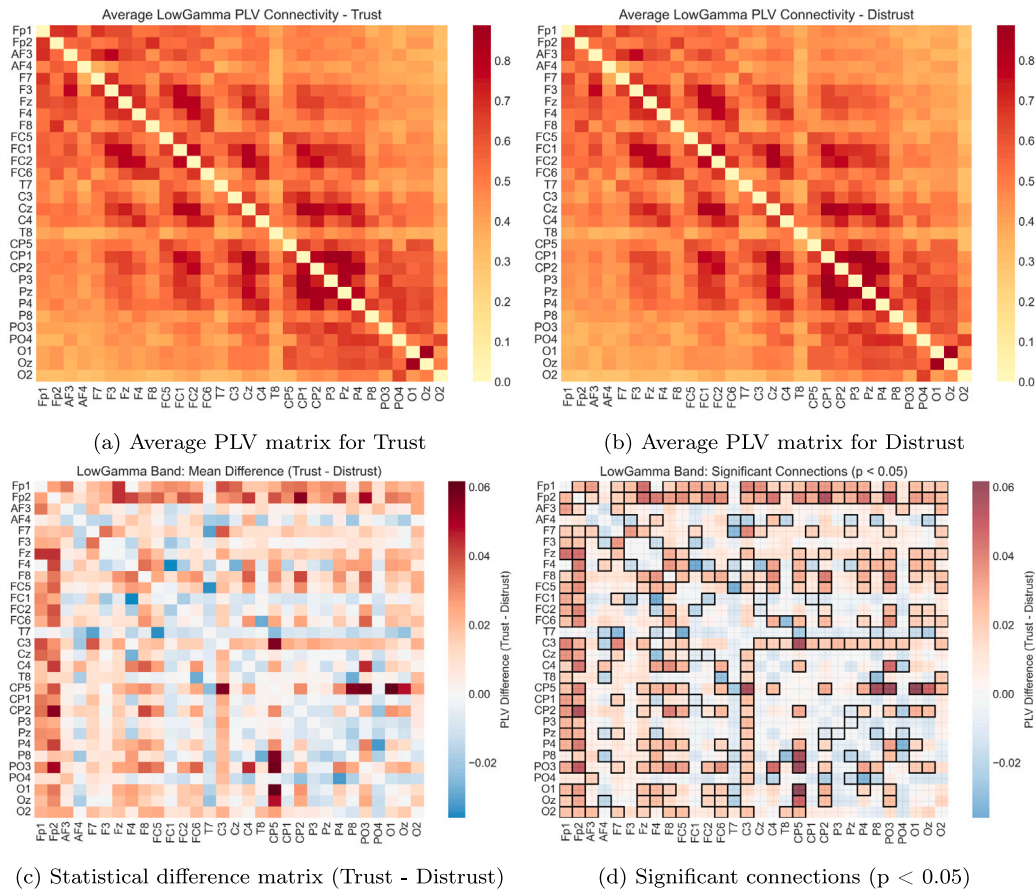


Fig. 6. Functional connectivity analysis of trust and distrust states in the Low Gamma band (30–47 Hz). (e) Average PLV connectivity pattern during trust states, showing enhanced high-frequency synchronisation. (f) Average PLV connectivity pattern during distrust states, demonstrating reduced gamma-band connectivity. (g) Statistical difference matrix highlighting connections that are significantly stronger (red) or weaker (blue) during trust compared to distrust. (h) Significant connections ($p < 0.05$) after thresholding by PLV strength, showing the most discriminative gamma-band network patterns for trust classification. **Colour scale:** Blue indicates lower PLV values, red indicates higher PLV values. In the statistical difference matrices, red indicates connections stronger during Trust than Distrust, and blue indicates connections weaker during Trust than Distrust.

- **CP5 as Central Hub:** Channel CP5 appears in 50% of the most stable features, indicating its role as a critical integration node in trust-related neural networks, connecting parietal regions with occipital and other parietal areas.
- **Gamma Band Dominance:** 60% of the most stable features are in high or low gamma bands, with an additional 20% in beta bands, indicating that high-frequency oscillations provide both discriminative power and cross-validation stability for trust classification.
- **Parietal-Occipital Network Emphasis:** 70% of stable features involve parietal (CP5, P8, PO3) or occipital (O1) regions, highlighting the importance of sensory integration and visuospatial processing in trust decisions.

A detailed inspection of both the stable features (Table 10) and the broader feature set (Fig. 8) reveals consistent neurocognitive patterns:

- **Parietal-Occipital Integration:** Features such as CP5_O1_HighGamma and CP5_O1_LowGamma dominate the stable feature set, suggesting robust connectivity between parietal association areas and occipital visual processing regions during trust evaluation.
- **Frontal-Parietal Coordination:** Connections like FP2_PO3_HighGamma and FP2_CP2_LowGamma indicate sustained communication between prefrontal executive regions and parietal integration areas, supporting complex social decision-making.
- **Multi-Frequency Processing:** The same anatomical connections (e.g., CP5-O1) appear across multiple frequency bands (High

Gamma, Low Gamma, Beta), suggesting that trust processing involves parallel oscillatory mechanisms at different temporal scales.

- **Right Hemisphere Preference:** Features involving FP2 (right prefrontal) appear more frequently than left-sided counterparts, potentially reflecting the specialised role of the right hemisphere in social cognition and emotional processing.

This multi-level analysis confirms that gamma-band oscillations in parietal-occipital networks provide both high discriminative power and exceptional stability for trust classification. The prominence of CP5 as a network hub suggests that trust decisions heavily rely on the integration of sensory information with spatial and social context processing.

While the group-level feature importance offers a broad understanding, individual participant saliency maps across all 12 participants (Fig. 9) reveal significant inter-subject variability in neural contributions. These maps demonstrate that, although there are shared neural mechanisms of trust, each individual's brain may rely on a uniquely configured set of networks.

Analysis of individual neural signatures across the 12 participants reveals three distinct patterns of trust processing:

1. **Parietal-Occipital Dominant Pattern:** Participants 01, 03, 06, and 09 show the strongest saliency in posterior regions, particularly around CP5, O1, and PO3 channels. This pattern aligns with the group-level stable features and suggests reliance on sensory integration and visuospatial processing for trust decisions.

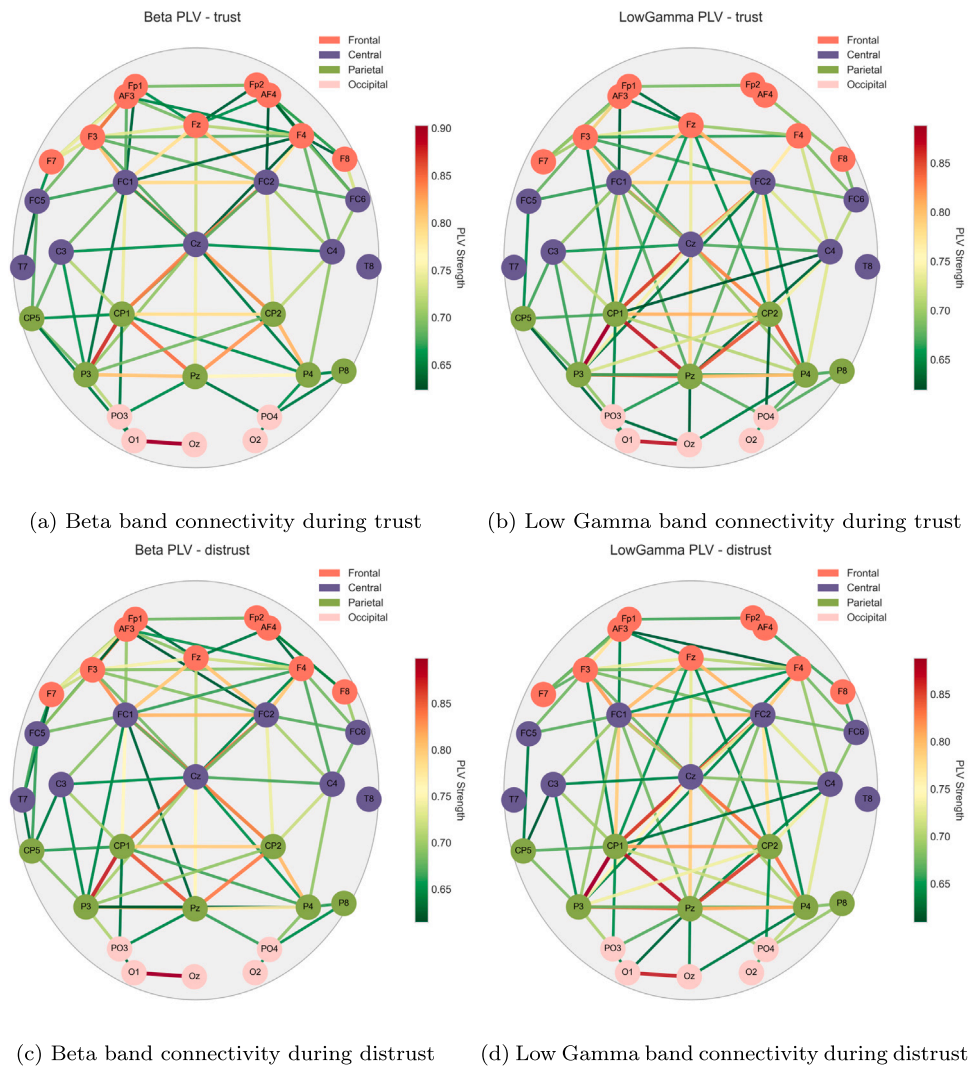


Fig. 7. PLV connectivity diagrams showing differential network patterns during trust and distrust states across frequency bands. (a) Trust in the Beta band exhibits widespread fronto-parietal and fronto-occipital synchronisation, supporting integrated cognitive processing. (b) Trust in the Low Gamma band shows enhanced high-frequency connectivity across distributed networks, indicating coordinated neural communication. (c) Distrust in the Beta band demonstrates reduced long-range connectivity and fragmented temporal-parietal pathways, reflecting disrupted network integration. (d) Distrust in the Low Gamma band shows sparse connectivity patterns, particularly in temporal and central regions, indicating impaired high-frequency coordination.

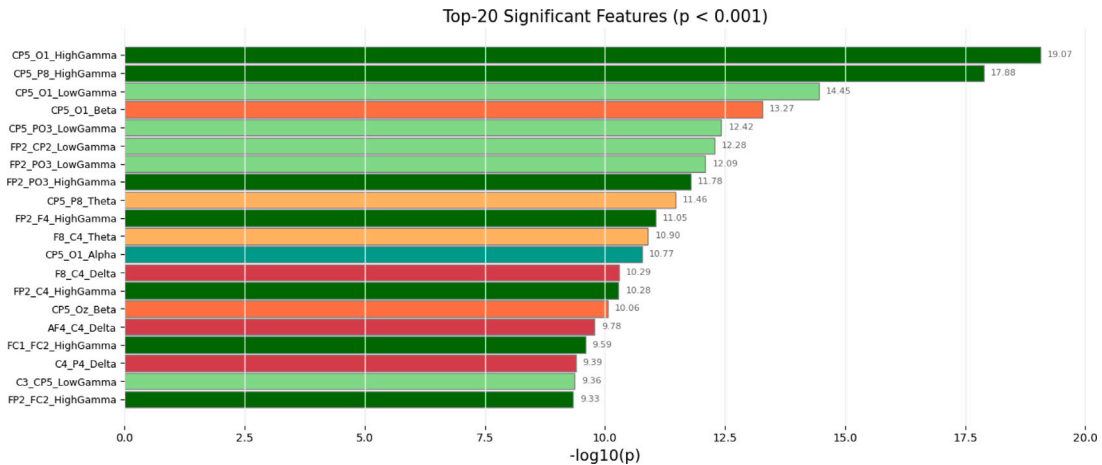


Fig. 8. Top 20 PLV connectivity features contributing most to trust classification. Channel pairs and frequency bands are colour-coded by band, spanning frontal, central, parietal, and occipital regions. High and low gamma bands are prominently represented, reflecting coordinated oscillatory activity associated with trust-related cognitive processes. See Table 10 for features demonstrating 100% selection stability across cross-validation folds.

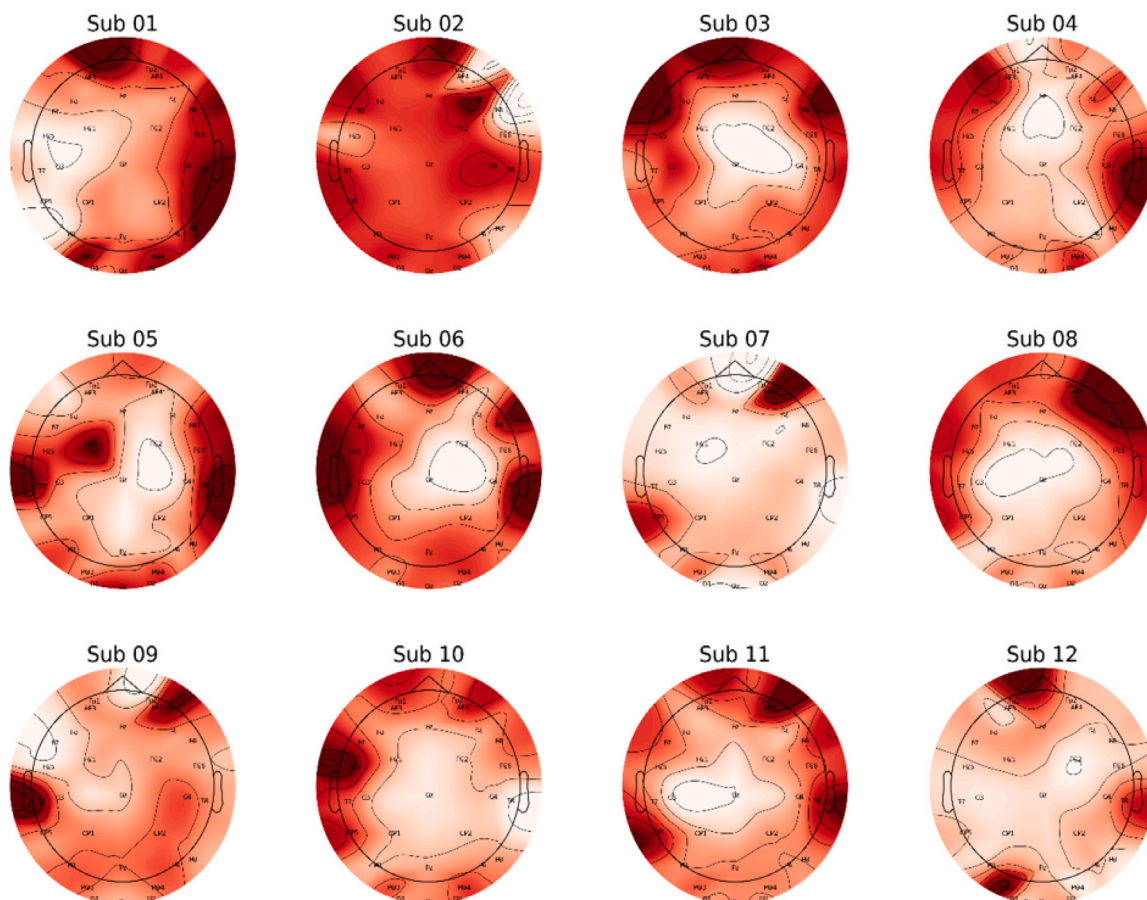


Fig. 9. Individual saliency maps for the PLV feature-based trust classification model across 12 participants (Subjects 01–12).

2. **Frontal-Executive Pattern:** Participants 02, 04, 08, and 11 demonstrate prominent frontal activation, with strong saliency in FP2, F4, and FC2 channels. This pattern indicates greater reliance on executive control and cognitive evaluation during trust formation.
3. **Distributed Network Pattern:** Participants 05, 07, 10, and 12 show more balanced activation across frontal, parietal, and occipital regions, suggesting integrated processing across multiple neural systems for trust evaluation.

Notable individual variations include:

- **Subject 07** shows unique right-hemisphere dominance, particularly in frontal-parietal connections
- **Subject 10** exhibits unusually strong occipital activation, potentially indicating heightened visual processing of trust cues
- **Subjects 04 and 11** show complementary frontal patterns, with Subject 04 emphasising medial frontal regions and Subject 11 showing more lateral prefrontal engagement

This variability underscores the need to consider individual differences in neural markers of trust, especially when developing personalised trust assessment or neurofeedback systems. The consistent appearance of CP5 and parietal-occipital connections across multiple participants (particularly in Pattern 1) provides validation for the group-level stable features identified in Table 10.

Together, these group-level and individual analyses reveal that trust-related cognitive states emerge from complex, distributed, and dynamically interacting brain networks. The stable parietal-occipital gamma-band connectivity (Table 10) represents core trust mechanisms involving sensory integration and contextual processing, while the individual variability patterns (Fig. 9) reflect the multi-dimensional nature

of trust processing across different cognitive strategies and individual neurocognitive profiles.

4. Discussion

The findings of this study provide compelling evidence for the utility of PLV-based functional connectivity in quantifying trust and distrust states in HAI. To ensure a coherent interpretation of our results, the discussion is structured around the three research questions (RQs), followed by broader implications and future directions.

First, addressing **RQ1**, the proposed PLVNet model reliably learned discriminative representations from PLV matrices, achieving superior classification performance compared to CNN, SVM, and KNN across both aggregated and participant-wise analyses (Tables 1, 5). This advantage stems from PLVNet's fully connected architecture, which is inherently suited for modelling arbitrary pair-wise relationships between EEG channels. By capturing subtle, non-linear synchronisation patterns that are typically overlooked in power-based or simplistic connectivity models, PLVNet provides robust, generalisable performance even across participants with distinct neural signatures. These results establish PLVNet as an effective and interpretable DL framework for trust classification.

For **RQ2**, PLV-based functional connectivity features revealed distinct neural signatures differentiating trust and distrust states. Trust was characterised by enhanced global integration, particularly across fronto-parietal and fronto-occipital pathways, reflecting increased co-ordination between executive control and sensory integration regions. Conversely, distrust exhibited fragmented synchronisation, especially within temporal-parietal pathways, suggesting diminished coherence

between regions supporting social reasoning and uncertainty monitoring. Notably, hub-like connections involving CP1, Pz, CP2, and Cz remained consistently dominant across both states and frequency bands, indicating a stable core network architecture that flexibly reorganises depending on the cognitive context. These findings link trust-related network integration to broader cognitive processes: increased fronto-parietal synchronisation aligns with sustained attention and top-down control [33], while reduced temporal-parietal connectivity during distrust overlaps with regions implicated in mentalising and perspective-taking [11]. Together, these results provide a mechanistic explanation for how distributed cortical networks support trust dynamics.

Turning to **RQ3**, the frequency-specific analyses revealed that the Beta and Low-Gamma bands were the most informative neural markers of trust-related processes (Table 9). Low-Gamma connectivity provided the highest classification accuracy, with trust states exhibiting widespread synchronisation between frontal, parietal, and occipital regions. Gamma-band activity has been associated with feature binding and dynamic integration [34], suggesting that trust relies on the rapid coupling of distributed cortical assemblies for evaluating partner reliability. The Beta band, while slightly less predictive, also contributed meaningfully to trust discrimination, consistent with its established role in maintaining cognitive stability and top-down attentional control [35,36]. These findings highlight a complementary role: Beta supports maintaining ongoing cognitive sets during stable trust, while Low-Gamma facilitates rapid network reconfiguration during trust evaluation.

For **RQ4**, the feature importance analysis (Fig. 8) and stability validation (Table 10) collectively demonstrate that trust engages distributed neural networks, while the individual variability patterns (Fig. 9) reveal substantial heterogeneity in how these networks are configured across participants. This multi-level evidence suggests that trust computation involves both shared neural substrates and individually tailored network engagement, supporting the development of personalised trust assessment approaches that can accommodate neurocognitive diversity while maintaining generalisable biomarkers.

Finally, from an applied perspective, the participant-wise consistency of PLVNet's performance (Fig. 4) demonstrates its potential for real-time neuroadaptive systems. By leveraging stable PLV-based features, such systems could dynamically monitor user trust and adjust interface feedback, automation levels, or decision support strategies to enhance performance and safety in HAI contexts.

4.1. Limitation and future research direction

Despite the contributions of this study, several limitations should be considered. The sample size was relatively small (12 participants) and homogeneous (all male), which may limit generalisability and increase the risk of overfitting. Small and uniform samples can reduce the reliability of machine learning predictions and restrict applicability to broader populations. Additionally, the study relied on univariate Mann-Whitney U tests for feature selection, which, while preserving a rich set of discriminative PLV connectivity patterns (particularly in frontal, parietal, and fronto-central regions), may not fully capture multivariate interactions or higher-order dependencies among features. Finally, the analysis focused on single-band connectivity, leaving cross-frequency interactions and causal relationships unexplored.

Future work should aim to validate the models on larger and more diverse cohorts, including different genders, age groups, and cultural backgrounds, to enhance generalisability. Exploring multivariate or embedded feature selection techniques (e.g., RFE, LASSO, or deep learning-based methods) could optimise feature dimensionality and improve model performance while maintaining interpretability. Investigating cross-frequency interactions (e.g., theta-gamma coupling) and extending analyses to effective connectivity will help uncover hierarchical neural mechanisms and causal information flow underlying trust. Integrating these approaches will strengthen the robustness, reliability, and real-world applicability of EEG-based trust monitoring in safety-critical contexts.

5. Conclusion

This study presents a novel framework for quantifying trust in HAI by leveraging PLV-based functional connectivity features combined with a lightweight DL model, PLVNet. To the best of our knowledge, this is the first study to investigate the use of PLV-based EEG functional connectivity features for real-time trust classification, offering insights into the distributed neural dynamics underlying trust and distrust states. By leveraging network-level EEG representations, the proposed approach demonstrates the feasibility of real-time cognitive state assessment, with participant-independent generalisability validated through leave-one-subject-out cross-validation. Statistical analyses confirmed that PLVNet significantly outperforms conventional benchmark models across multiple performance metrics, underscoring the robustness and reliability of the approach.

Careful feature selection using Mann-Whitney U tests preserved reproducible and interpretable connectivity patterns, with group-level analysis revealing distributed cortical networks and stability validation confirming robust parietal-occipital gamma-band connectivity as core trust biomarkers. Individual variability patterns further demonstrated that while trust engages common neural substrates, each individual exhibits unique network configurations, highlighting the importance of personalised approaches in trust assessment.

Beyond classification performance, the findings reveal the critical role of Beta and Low-Gamma frequency bands in supporting rapid evaluative and decision-making processes during trust formation. These insights provide a neurophysiological foundation for developing targeted monitoring strategies and designing interventions that enhance operator-automation collaboration, particularly through adaptive systems that can accommodate individual neurocognitive differences.

In summary, this work integrates advanced PLV-based connectivity analysis with a purpose-built DL framework to decode trust states in HAI. By revealing how trust manifests as enhanced orchestration across distributed cortical networks and distrust as fragmented synchronisation, we provide both a validated neurophysiological account and a practical, interpretable model for real-time trust detection. The demonstrated individual variability in neural trust signatures offers clear pathways for future work involving personalised trust modelling, larger cohorts, multi-frequency connectivity, and adaptive neuro-aware automation systems that respect and leverage human neurodiversity.

CRedit authorship contribution statement

Julakha Jahan Jui: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Imali T. Hettiarachchi:** Conceptualization, Supervision, Visualization, Writing – review & editing. **Asim Bhatti:** Supervision, Visualization, Writing – review & editing. **Douglas Creighton:** Visualization, Writing – review & editing.

Ethics statement

The proposed research satisfies the conditions of 'low-risk research' according to the National Statement on Ethical Conduct in Human Research 2007. A pre-assessment has been completed, and a low-risk ethics application has already been approved by the Human Research Ethics Advisory Group (HEAG) of the Faculty of Science and Technology, Deakin University (Reference Number: SEBE-2024-17).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research is fully funded by the Institute for Intelligent Systems Research and Innovation (IISRI), Deakin University, Australia.

References

- [1] J.D. Lee, K.A. See, Trust in automation: Designing for appropriate reliance, *Hum. Factors* 46 (1) (2004) 50–80.
- [2] K.A. Hoff, M. Bashir, Trust in automation: Integrating empirical evidence on factors that influence trust, *Hum. Factors* 57 (3) (2015) 407–434.
- [3] J.J. Jui, I.T. Hettiarachchi, N. Mohajer, Need for trust calibration in takeover request performance in level 3 automated vehicles, in: *Proceedings of the 17th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 2025, pp. 296–306.
- [4] M.T. Dzindolet, S.A. Peterson, R.A. Pomranky, L.G. Pierce, H.P. Beck, The role of trust in automation reliance, *Int. J. Hum.-Comput. Stud.* 58 (6) (2003) 697–718.
- [5] T. Zhang, J. Yang, M. Chen, Z. Li, J. Zang, X. Qu, EEG-based assessment of driver trust in automated vehicles, *Expert Syst. Appl.* 246 (2024) 123196.
- [6] M. Seet, J. Harvy, R. Bose, A. Dragomir, A. Bezerianos, N. Thakor, Differential impact of autonomous vehicle malfunctions on human trust, *IEEE Trans. Intell. Transp. Syst.* 23 (1) (2020) 548–557.
- [7] J.J. Jui, I.T. Hettiarachchi, A. Bhatti, M.R.M. Farghaly, D. Creighton, A recent review on subjective and objective assessment of trust in human autonomy teaming, *IEEE Trans. Human-Mach. Syst.* (2025) 819–833.
- [8] J.-Y. Jian, A.M. Bisantz, C.G. Drury, Foundations for an empirically determined scale of trust in automated systems, *Int. J. Cogn. Ergon.* 4 (1) (2000) 53–71.
- [9] Y. Zhang, A. Yadav, S.K. Hopko, R.K. Mehta, In gaze we trust: Comparing eye tracking, self-report, and physiological indicators of dynamic trust during HRI, in: *Companion of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*, 2024, pp. 1188–1193.
- [10] J.J. Jui, I.T. Hettiarachchi, A. Bhatti, M.R.M. Farghaly, Real-time assessment of trust using EEG and artificial neural network, in: *Proceedings of the International Joint Conference on Neural Networks, IJCNN*, 2025, Accepted.
- [11] P. Walla, S. Kalt, D. Külzer, Even-related potentials (ERPs) reveal that trust and distrust differ between brands and political institutions, in: *NeuroIS Retreat*, Springer, 2023, pp. 1–7.
- [12] C. Blais, D.M. Ellis, K.M. Wingert, A.B. Cohen, G.A. Brewer, Alpha suppression over parietal electrode sites predicts decisions to trust, *Soc. Neurosci.* 14 (2) (2019) 226–235.
- [13] J. Huang, S. Choo, Z.H. Pugh, C.S. Nam, Evaluating effective connectivity of trust in human-automation interaction: A dynamic causal modeling (DCM) study, *Hum. Factors* 64 (6) (2022) 1051–1069.
- [14] P. Sauseng, W. Klimesch, M. Schabus, M. Doppelmayr, Fronto-parietal EEG coherence in theta and upper alpha reflect central executive functions of working memory, *Int. J. Psychophysiol.* 57 (2) (2005) 97–103.
- [15] G.M. Clements, D.C. Bowie, M. Gyurkovics, K.A. Low, M. Fabiani, G. Gratton, Spontaneous alpha and theta oscillations are related to complementary aspects of cognitive control in younger and older adults, *Front. Hum. Neurosci.* 15 (2021) 621620.
- [16] J.F. Cavanagh, A.J. Shackman, Frontal midline theta reflects anxiety and cognitive control: meta-analytic evidence, *J. Physiol.-Paris* 109 (1–3) (2015) 3–15.
- [17] H. Chen, I.Y. Chan, Z. Dong, T.-A. Samuel, Unraveling trust in collaborative human-machine intelligence from neurophysiological perspective: a review of EEG and fNIRS features, *Adv. Eng. Inform.* 67 (2025) 103555.
- [18] F. Al-Shargie, Y. Badr, E. Dahy, U. Tariq, H. Al-Nashash, Classification of four mental states using EEG connectivity and convolutional neural networks, *IEEE Sensors J.* (2025).
- [19] A. Hag, D. Handayani, T. Pillai, T. Mantoro, M.H. Kit, F. Al-Shargie, EEG mental stress assessment using hybrid multi-domain feature sets of functional connectivity network and time-frequency features, *Sensors* 21 (18) (2021) 6300.
- [20] M.-a. Talaat, M. Awadalla, L. Abdel-Hamid, Recurrence quantification analysis (RQA) features vs. Traditional EEG features for alzheimer's disease diagnosis, *Intel. Artif.* 28 (75) (2025) 170–185.
- [21] A. Cacciotti, C. Pappalettera, F. Miraglia, P.M. Rossini, F. Vecchio, EEG entropy insights in the context of physiological aging and Alzheimer's and Parkinson's diseases: A comprehensive review, *GeroScience* 46 (6) (2024) 5537–5557.
- [22] K. Akash, W.-L. Hu, N. Jain, T. Reid, A classification model for sensing human trust in machines using EEG and GSR, *ACM Trans. Interact. Intell. Syst. (TiiS)* 8 (4) (2018) 1–20.
- [23] G. Bales, A. Hayman, T. Clark, J. Dekarske, S. Joshi, Z. Kong, An EEG-network-metric based approach to real-time trust inference in human-autonomy teaming, *Front. Neuroergon.* 6, 1627483.
- [24] K.F. Firoz, Y. Seong, Y. Chang, A preliminary study on human trust in pseudo-real-time scenario through electroencephalography and machine learning based data classification, in: *1st International Conference on Smart Mobility and Vehicle Electrification*, 2023.
- [25] C. Berka, D.J. Levendowski, M.M. Cvetinovic, M.M. Petrovic, G. Davis, M.N. Lumicao, V.T. Zivkovic, M.V. Popovic, R. Olmstead, Real-time analysis of EEG indexes of alertness, cognition, and memory acquired with a wireless EEG headset, *Int. J. Hum.-Comput. Interact.* 17 (2) (2004) 151–170.
- [26] S. Oh, Y. Seong, S. Yi, S. Park, Investigation of human trust by identifying stimulated brain regions using electroencephalogram, *ICT Express* 8 (3) (2022) 363–370.
- [27] A. Delorme, S. Makeig, EEGLAB: an open source toolbox for analysis of single-trial EEG dynamics including independent component analysis, *J. Neurosci. Methods* 134 (1) (2004) 9–21.
- [28] N. Bajaj, J.R. Carrión, F. Bellotti, R. Berta, A. De Gloria, Automatic and tunable algorithm for EEG artifact removal using wavelet decomposition with applications in predictive modeling during auditory tasks, *Biomed. Signal Process. Control.* 55 (2020) 101624.
- [29] H.B. Mann, D.R. Whitney, On a test of whether one of two random variables is stochastically larger than the other, *Ann. Math. Stat.* (1947) 50–60.
- [30] J.J. Jui, I.T. Hettiarachchi, Temporal convolutional networks for driver fatigue classification from EEG signals: A novel approach, in: *IEEE International Conference on Systems, Man, and Cybernetics, SMC*, 2025, Accepted.
- [31] J. Demšar, Statistical comparisons of classifiers over multiple data sets, *J. Mach. Learn. Res.* 7 (Jan) (2006) 1–30.
- [32] C. Fu, Z. Zhang, J. He, S. Huang, J. Qiu, Y. Wang, Brain dynamics of decision-making in the generalized trust game: Evidence from ERPs and EEG time-frequency analysis, *Acta Psychol. Sin.* (2018).
- [33] Q. Chenot, E. Lepron, X. De Boissezon, S. Scannella, Functional connectivity within the fronto-parietal network predicts complex task performance: A fNIRS study, *Front. Neuroergon.* 2 (2021) 718176.
- [34] T. Ossandón, K. Jerbi, J.R. Vidal, D.J. Bayle, M.-A. Henaff, J. Jung, L. Minotti, O. Bertrand, P. Kahane, J.-P. Lachaux, Transient suppression of broadband gamma power in the default-mode network is correlated with task complexity and subject performance, *J. Neurosci.* 31 (41) (2011) 14521–14530.
- [35] J. Kamiński, A. Brzezicka, M. Gola, A. Wróbel, Beta band oscillations engagement in human alertness process, *Int. J. Psychophysiol.* 85 (1) (2012) 125–128.
- [36] V. Betti, S. Della Penna, F. de Pasquale, M. Corbetta, Spontaneous beta band rhythms in the predictive coding of natural stimuli, *Neurosci.* 27 (2) (2021) 184–201.